Credal Uncertainty Quantification for Reliable Stance Detection on Unseen Targets

Abstract

Stance detection identifies whether the author of a text is in *favor* of, *against*, or *neutral* towards a given target [1, 3]. Current models, from fine-tuned transformers to large language models (LLMs), output a single probability distribution over stance labels. That one distribution mixes up two different sources of uncertainty [8]: *aleatoric* uncertainty, which comes from sarcasm, implicit stance, missing context and genuine disagreement between human annotators [4, 15], and *epistemic* uncertainty, which comes from targets, domains and events that the model has never seen. As a result, stance detectors are often over-confident on unseen targets and cannot give a principled reason to abstain.

We propose CrediStance, a stance detection framework based on *credal sets*, i.e. convex sets of probability distributions [6]. Instead of a single point estimate, CrediStance predicts a lower and an upper probability for each stance. It builds credal sets from diverse ensembles of transformer encoders and LoRA-adapted LLMs [10, 11, 13], and splits total uncertainty into aleatoric and epistemic parts using upper and lower entropy [7]. It then calibrates the credal sets with conformal prediction, giving a distribution-free guarantee that the human label distribution is covered [12]. Finally, it applies imprecise-probability decision rules [9] to output a single stance, a set of plausible stances, or a deferral to a human reviewer.

The project collects *no new annotations*. Human label distributions come from the publicly released raw annotator labels of SemEval-2016, with about eight independent annotators per tweet for 4,306 tweets [1, 2]. From these we build three human-grounded evaluation settings: (i) an in-target test set, (ii) an *unseen-target* test set, obtained by holding out an entire target during training, and (iii) an *ambiguity* stress test of 1,385 low-agreement tweets that were excluded from the original shared task. ChaosNLI, which has about 100 annotators per item, serves as a further check of conformal coverage [14]. We also evaluate zero-shot stance detection on VAST [4] and EZ-Stance [5]. Baselines are softmax, temperature scaling, Monte Carlo dropout, deep ensembles, evidential deep learning, conformal prediction and LLM confidence. We report calibration, selective-prediction risk, unseen-target detection, conformal coverage, and correlation between predicted aleatoric uncertainty and human disagreement. To our knowledge, this is the first application of credal uncertainty quantification to stance detection. It aims to make stance classification more trustworthy for content moderation, misinformation analysis and opinion mining.

Keywords: stance detection, uncertainty quantification, credal sets, imprecise probabilities, aleatoric and epistemic uncertainty, conformal prediction, zero-shot learning, human label variation, large language models.

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